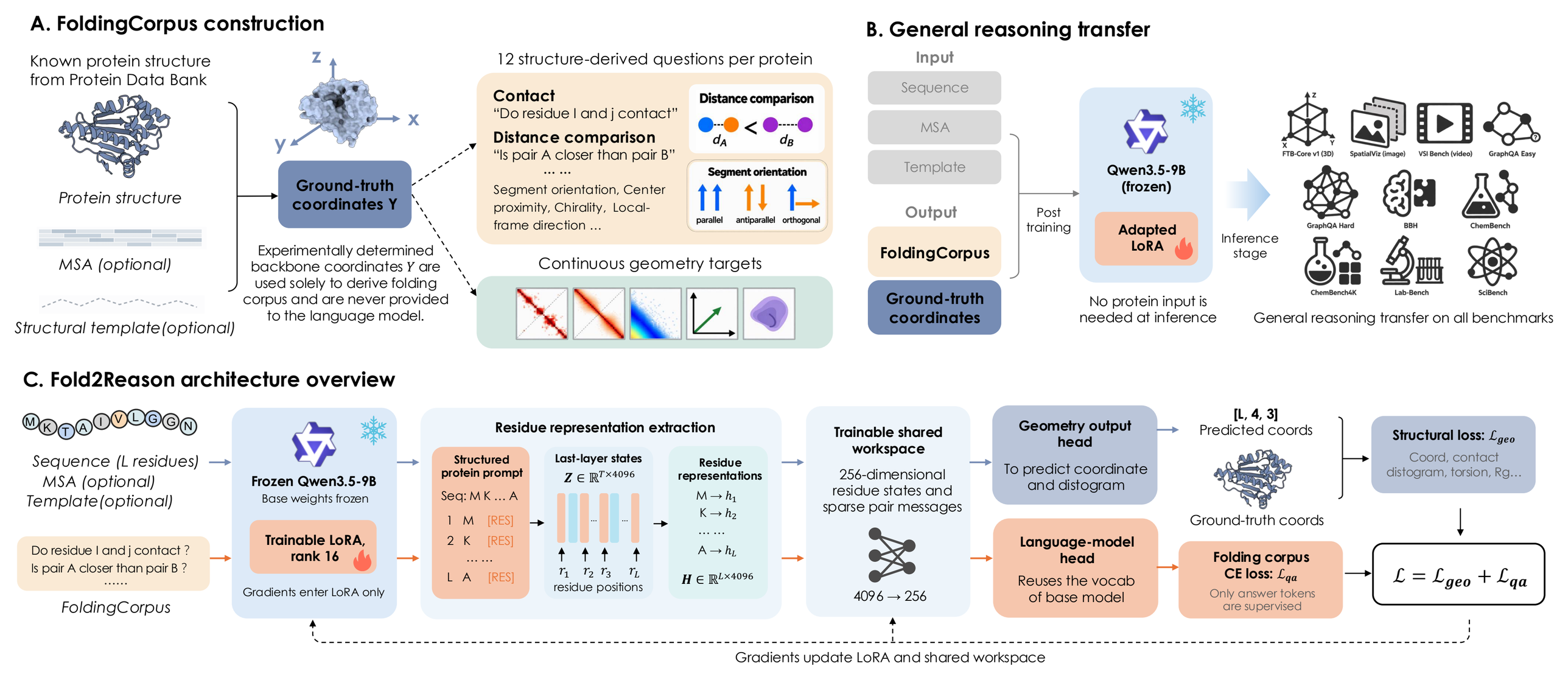

A. FoldingCorpus construction
Known protein structure from Protein Data Bank
Protein structure
MSA (optional)
Structural template(optional)
z
x
y
Ground-truth coordinates Y
Experimentally determined backbone coordinates Y are used solely to derive folding corpus and are never provided to the language model.
12 structure-derived questions per protein
Contact
"Do residue I and j contact"
Distance comparison
"Is pair A closer than pair B"
… …
Segment orientation, Center proximity, Chirality, Local-frame direction …
Distance comparison
Segment orientation
parallel
antiparallel
orthogonal
Continuous geometry targets
B. General reasoning transfer
Input
Sequence
MSA
Template
Output
FoldingCorpus
Ground-truth coordinates
Post training
Qwen3.5-9B (frozen)
Adapted LoRA
No protein input is needed at inference
Inference stage
FTB-Core v1 (3D)
SpatialViz (image)
VSI Bench (video)
GraphQA Easy
GraphQA Hard
BBH
ChemBench
ChemBench4K
Lab-Bench
SciBench
General reasoning transfer on all benchmarks
C. Fold2Reason architecture overview
Sequence (L residues)
MSA (optional)
Template(optional)
Do residue I and j contact ?
Is pair A closer than pair B ?
……
FoldingCorpus
Frozen Qwen3.5-9B
Base weights frozen
Trainable LoRA, rank 16
Gradients enter LoRA only
Residue representation extraction
Structured protein prompt
Seq: M K … A
1 M [RES]
2 K [RES]
… …
L A [RES]
Last-layer states
$Z \in \mathbb{R}^{T\times 4096}$
residue positions
Residue representations
M → $h_1$
K → $h_2$
… …
A → $h_L$
$H \in \mathbb{R}^{L\times 4096}$
Trainable shared workspace
256-dimensional residue states and sparse pair messages
4096 → 256
Geometry output head
To predict coordinate and distogram
[L, 4, 3]
Predicted coords
Ground-truth coords
Structural loss: $\mathcal{L}_{geo}$
Coord, contact distogram, torsion, Rg…
Language-model head
Reuses the vocab of base model
Folding corpus CE loss: $\mathcal{L}_{qa}$
Only answer tokens are supervised
$\mathcal{L} = \mathcal{L}_{geo} + \mathcal{L}_{qa}$
Gradients update LoRA and shared workspace